

Intermediate

1. The equation of a curve is $y = (x - 3)\sqrt{2x + 1}$.

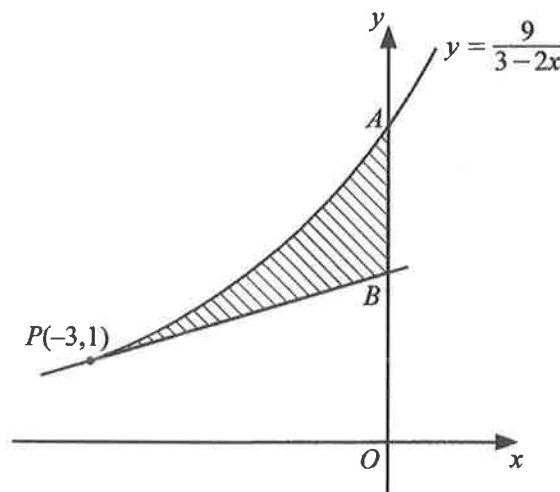
(a) Show that $\frac{dy}{dx} = \frac{3x - 2}{\sqrt{2x + 1}}$. [4]

(b) A particle moves along the curve in such a way that the x -coordinate is increasing at a constant rate of 0.15 units per second. Find the rate of change of the y -coordinate at the point where $x = 4$. [2]

(c) Use the result from **part (a)** to evaluate $\int_0^4 \frac{x}{\sqrt{2x + 1}} dx$. [4]

[N22/II/8]

2.



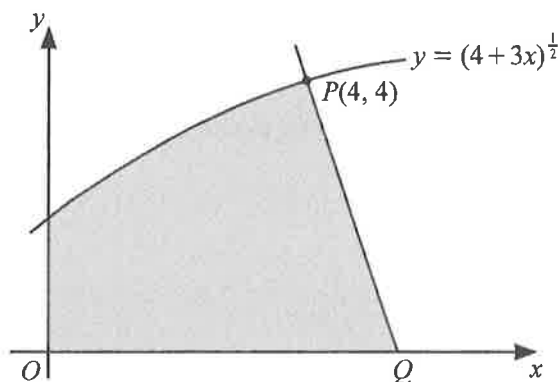
The diagram shows part of the curve $y = \frac{9}{3-2x}$ and the tangent to the curve at the point $P(-3, 1)$. The curve intersects the y -axis at A and the tangent intersects the y -axis at B .

(a) Determine, with full working, whether B is nearer to O or to A . [6]

(b) Find, showing all necessary working, the value of the constants a and b for which the area of the shaded region can be expressed as $a \ln 3 + b$. [6]

[N22/II/10]

3.



The diagram shows part of the curve $y = (4 + 3x)^{\frac{1}{2}}$. The point $P(4, 4)$ lies on the curve and the normal to the curve at P meets the x -axis at Q .

(a) Find the coordinates of Q . [5]

(b) Find the area of the shaded region bounded by the curve, the normal PQ and the coordinate axes. [5]

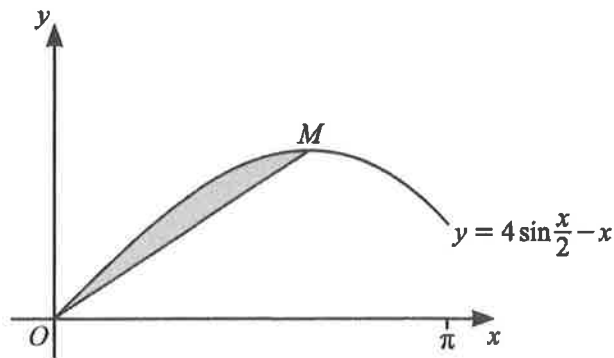
[N21/I/14]

4. (a) Given that $y = \frac{x}{(2x+1)^{\frac{1}{2}}}$, show that $\frac{dy}{dx} = \frac{x+1}{(2x+1)^{\frac{3}{2}}}$. [4]

(b) Hence find the value of $\int_0^4 \frac{x}{(2x+1)^{\frac{3}{2}}} dx$. [4]

[N21/II/3]

5.



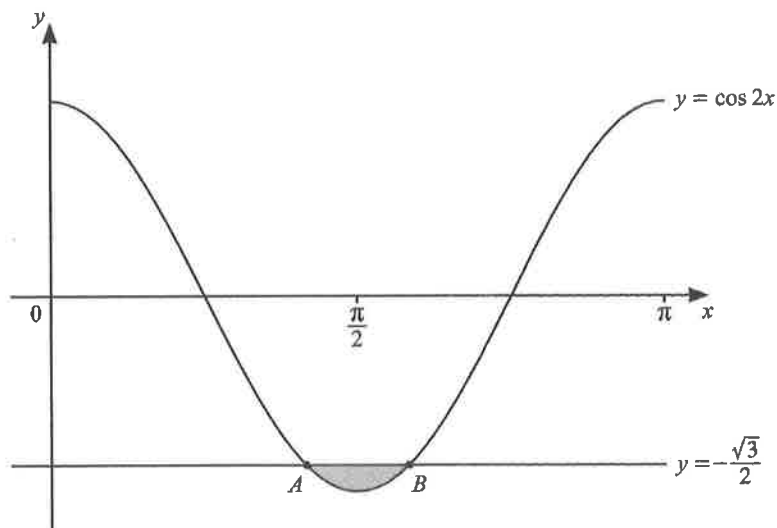
The diagram shows the curve $y = 4 \sin \frac{x}{2} - x$ for $0 \leq x \leq \pi$ radians. The point M is the maximum point of the curve and OM is a straight line.

Show that the area of the shaded region is $4 - \frac{2\pi\sqrt{3}}{3}$ units².

[12]

[N21/II/10]

6.



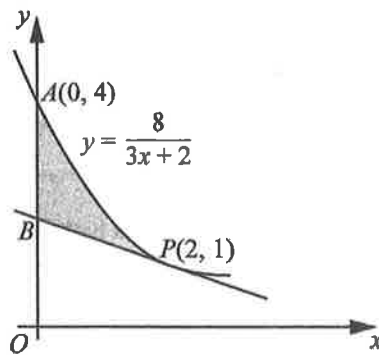
The diagram shows, for $0 \leq x \leq \pi$, the curve $y = \cos 2x$ and the line $y = -\frac{\sqrt{3}}{2}$. The curve and the line intersect at the points A and B .

(i) Find the x -coordinate of A and of B . [3]

(ii) Showing all your working, find the exact area of the shaded region. [6]

[N20/I/10]

7.

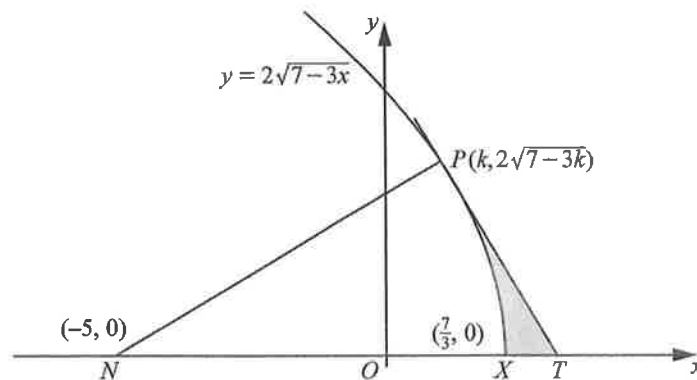


The diagram shows part of the curve $y = \frac{8}{3x+2}$ intersecting the y -axis at $A(0, 4)$. The tangent to the curve at the point $P(2, 1)$ intersects the y -axis at B . Find the exact area of the shaded region.

[11]

[N19/II/9]

8.



The diagram shows part of the curve $y = 2\sqrt{7-3x}$ passing through the point $P(k, 2\sqrt{7-3k})$, where k is a constant, $k < \frac{7}{3}$. The curve meets the x -axis at the point X . The tangent and normal to the curve at P meet the x -axis at the points T and $N(-5, 0)$ respectively.

(i) Prove that $k = 1$.

[5]

(ii) Show that the x -coordinate of T is $\frac{11}{3}$.

[2]

The curve meets the x -axis at the point $X\left(\frac{7}{3}, 0\right)$.

(iii) Find the area of the shaded region PXT .

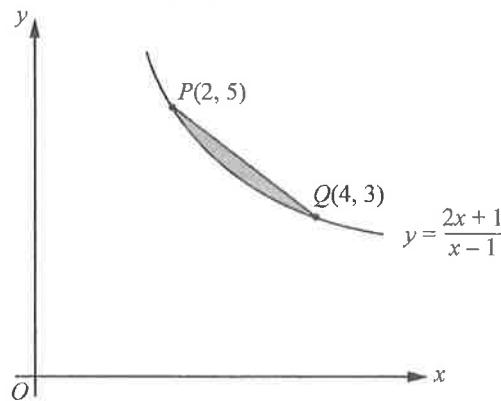
[5]

[N18/II/10]

9. The equation of a curve is $y = \frac{2x+1}{x-1}$.

(i) Explain, with working, why the curve has no turning points. [3]

The diagram below shows part of the curve $y = \frac{2x+1}{x-1}$. Points $P(2, 5)$ and $Q(4, 3)$ lie on the curve.



- (ii) By expressing $\frac{2x+1}{x-1}$ in the form $a + \frac{b}{x-1}$, where a and b are constants, find, showing full working, the area of the shaded region. [5]

[N17/I/11]

10. (i) Show that $\frac{d}{dx}(\tan x - x) = \tan^2 x$. [2]

(ii) Without using a calculator, find the value of each of the constants a and b for which

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sec^2 x + 5 \tan^2 x) dx = a\sqrt{3} + b\pi. \quad [5]$$

[N17/II/2]

11. It is given that $\int f(x) dx = \sin x + k \cos 2x + c$, where c is a constant of integration, and that $\int_0^{\frac{\pi}{6}} f(x) dx = \frac{3}{4}$.

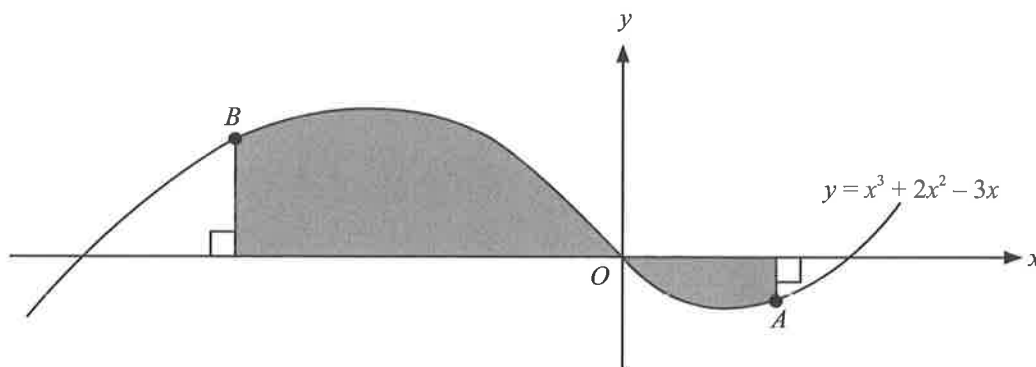
(i) Show that $k = -\frac{1}{2}$. [1]

(ii) Find $f(x)$. [3]

(iii) Find the equation of the normal to the curve $y = f(x)$ at the point where $x = \frac{\pi}{6}$. [5]

[N16/I/12]

12.



The diagram shows part of the graph of $y = x^3 + 2x^2 - 3x$. The x -coordinate of the point A is $\frac{2}{3}$.

(i) Find the gradient of the curve at A . [4]

The tangents to the curve at the points A and B are parallel.

(ii) Find the x -coordinate of B . [3]

(iii) Showing all your working, find the total area of the shaded region bounded by the curve, the x -axis and the lines from A and B perpendicular to the x -axis. [4]

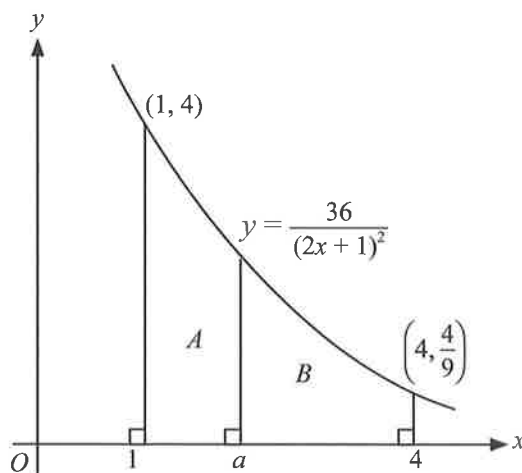
[N16/II/9]

13. The equation of a curve is $y = \frac{36}{(2x+1)^2}$.

(i) A point P moves along the curve in such a way that the x -coordinate of P increases at a constant rate of 0.02 units per second. Find the x -coordinate of P at the instant the y -coordinate is decreasing at a rate of 0.36 units per second. [5]

(ii) The diagram shows part of the curve $y = \frac{36}{(2x+1)^2}$ passing through the points with coordinates $(1, 4)$ and $(4, \frac{4}{9})$.

Also shown are lines perpendicular to the x -axis at the points with x -coordinates 1, a and 4.



Given that the areas of the regions marked A and B are equal, find the value of the constant a .

[5]

[N15/I/12]

14. (i) Show that $\frac{d}{dx}(\ln(\cos x)) = -\tan x$.

[2]

(ii) Differentiate $x \tan x$ with respect to x .

[2]

(iii) Using the results from parts (i) and (ii), find $\int x \sec^2 x \, dx$ and hence show that

$$\int_0^{\frac{\pi}{4}} x \sec^2 x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2.$$

[4]

[N15/II/2]

15. (i) Show that $\frac{d}{dx} \left\{ x(3x-5)^{\frac{5}{3}} \right\} = (8x-5)(3x-5)^{\frac{2}{3}}$. [5]
- (ii) Hence find $\int x(3x-5)^{\frac{2}{3}} dx$, giving your answer in the form $k(x+1)(3x-5)^{\frac{5}{3}} + c$, where k is a constant to be found and c is a constant of integration which cannot be found. [4]
- (iii) Find the value of $\int_{-1}^{\frac{5}{3}} x(3x-5)^{\frac{2}{3}} dx$ and explain what the result implies about the curve $y = x(3x-5)^{\frac{2}{3}}$. [2]
- [N20/II/7]